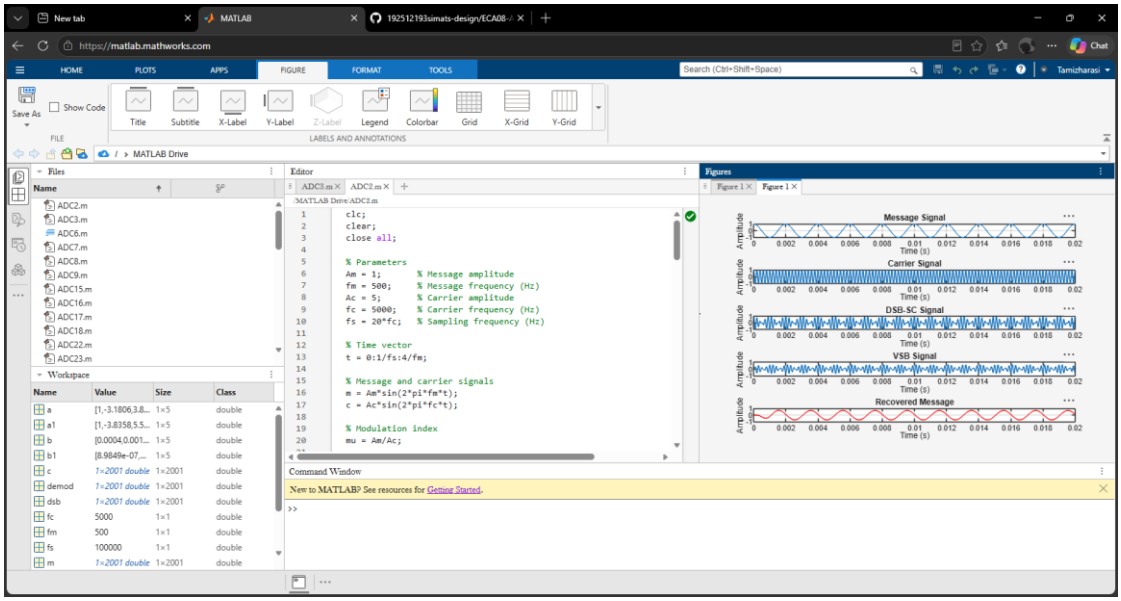
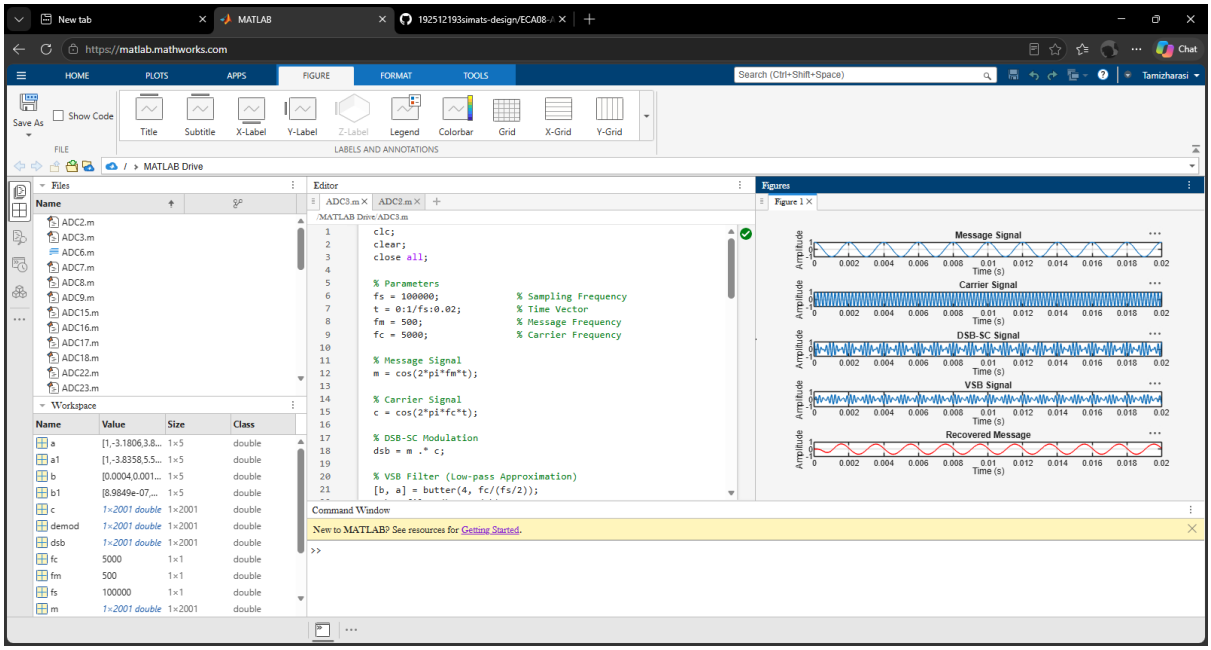
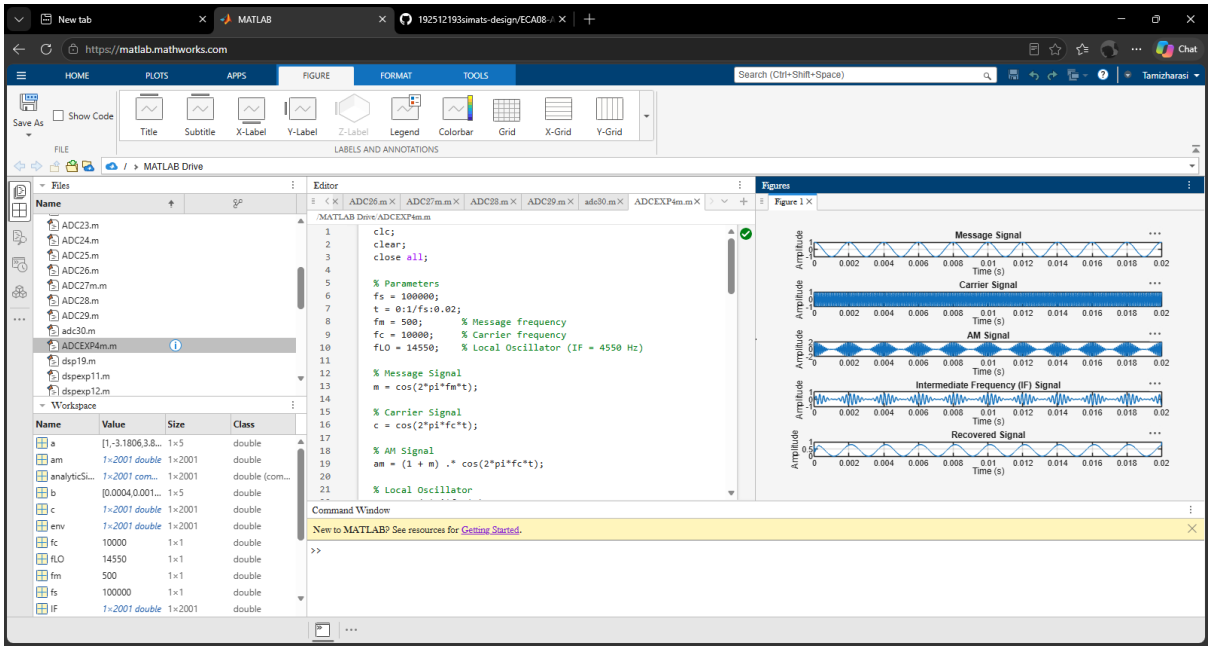
EXP 2


Message Signal
Carrier Signal
DSB-SC Signal
VSB Signal
Recovered Message
Amplitude
Time (s)

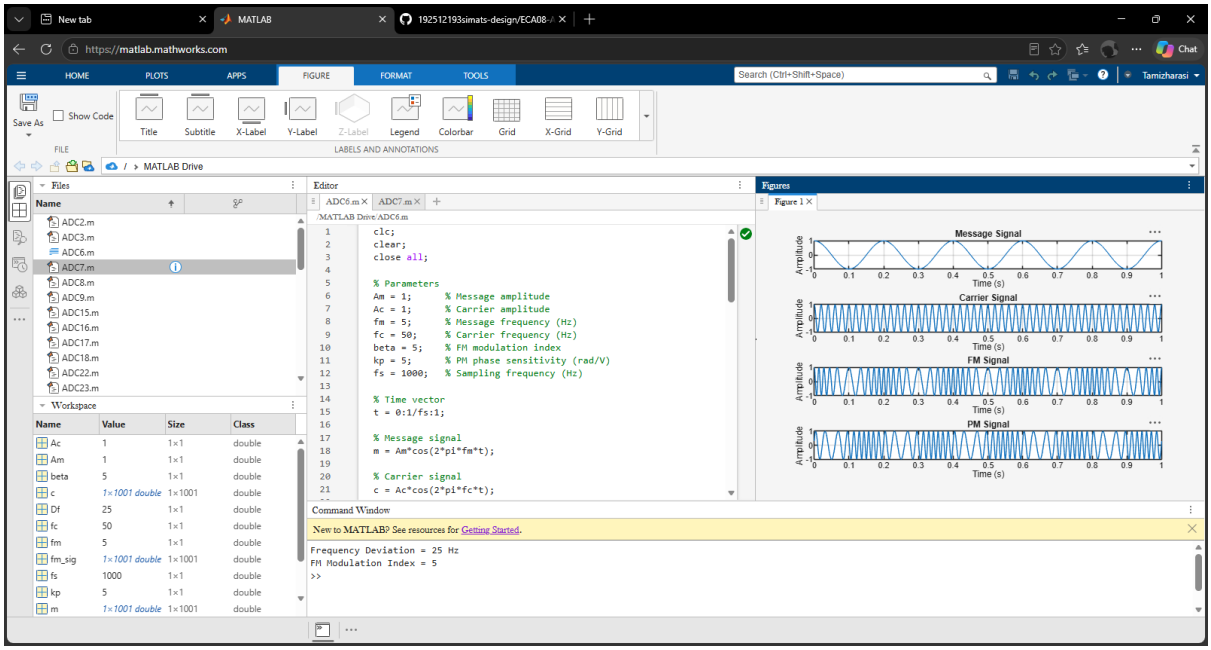
EXP3


Message Signal
Carrier Signal
DSB-SC Signal
VSB Signal
Recovered Message
Amplitude
Time (s)

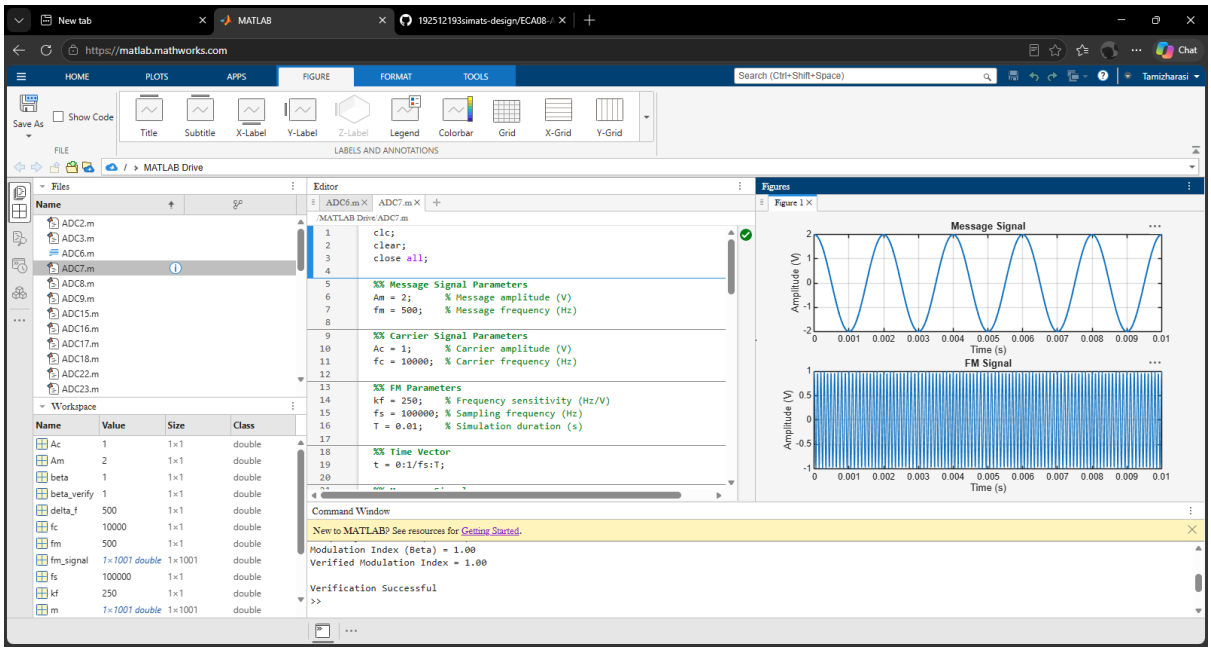
EXP4


Message Signal
Carrier Signal
AM Signal
Intermediate Frequency (IF) Signal
Recovered Signal
Amplitude
Time (s)

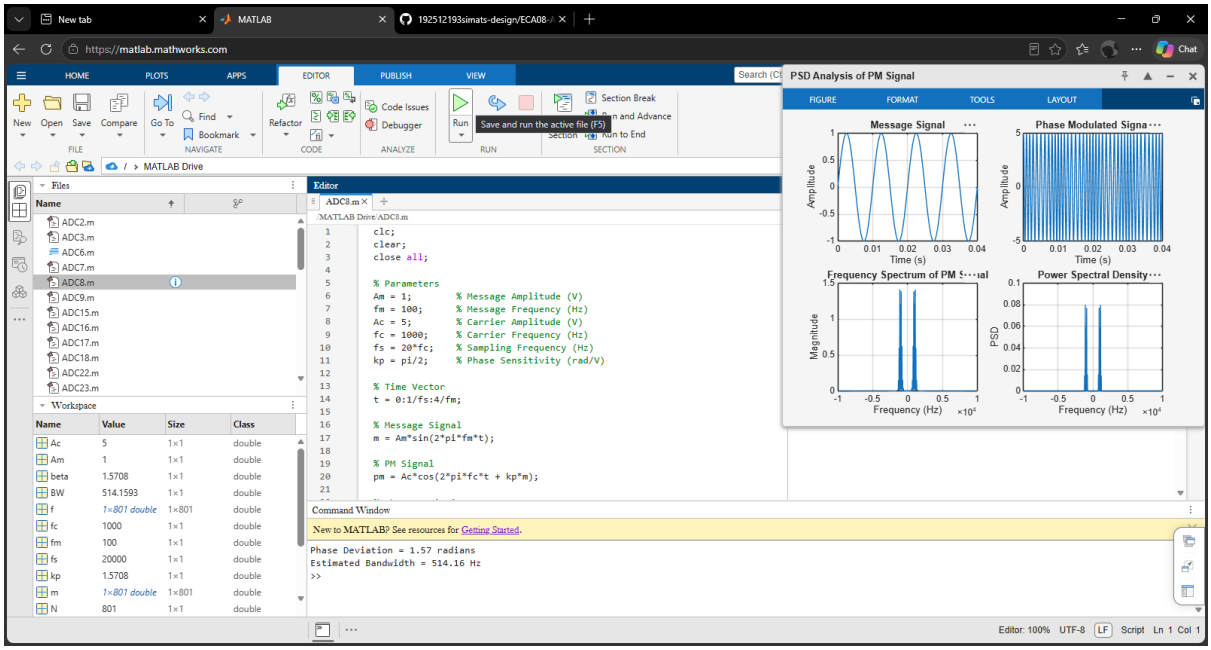
EXP 6


Message Signal
Carrier Signal
FM Signal
PM Signal
Amplitude
Time (s)

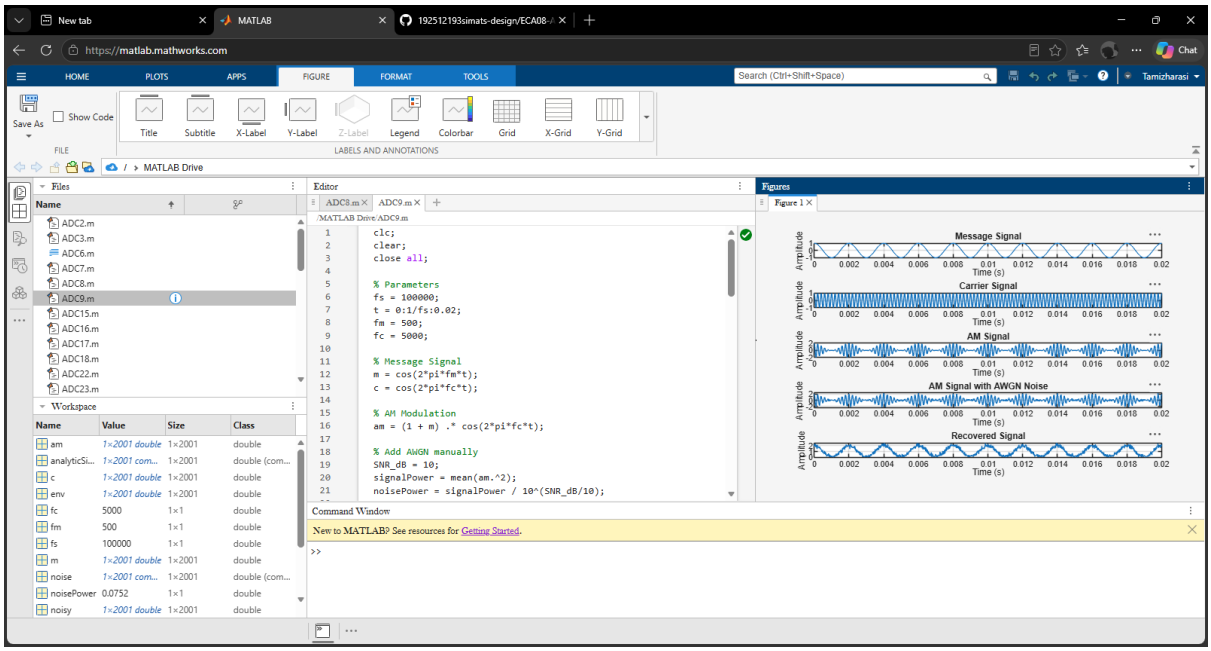
EXP7


Message Signal
FM Signal
Time (s)
Amplitude (V)
clc;
clear;
close all;
%% Message Signal Parameters
Am = 2; % Message amplitude (V)
fm = 500; % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)
%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.01; % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
Modulation Index (Beta) = 1.00
Verified Modulation Index = 1.00
Verification Successful

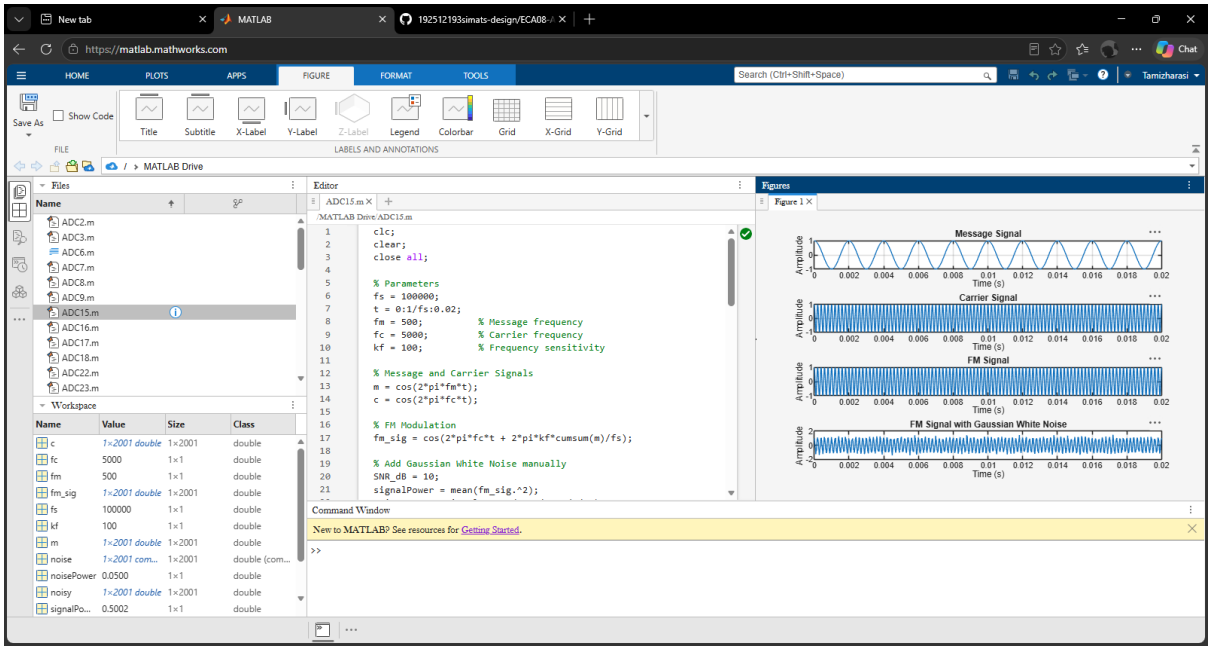
EXP8


PSD Analysis of PM Signal
Message Signal
Phase Modulated Signa
Frequency Spectrum of PM S
Power Spectral Density
Time (s)
Frequency (Hz)
Amplitude
Magnitude
PSD
clc;
clear;
close all;
% Parameters
Am = 1; % Message Amplitude (V)
fm = 100; % Message Frequency (Hz)
Ac = 5; % Carrier Amplitude (V)
fc = 1000; % Carrier Frequency (Hz)
fs = 20*fc; % Sampling Frequency (Hz)
kp = pi/2; % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% PM Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
Phase Deviation = 1.57 radians
Estimated Bandwidth = 514.16 Hz

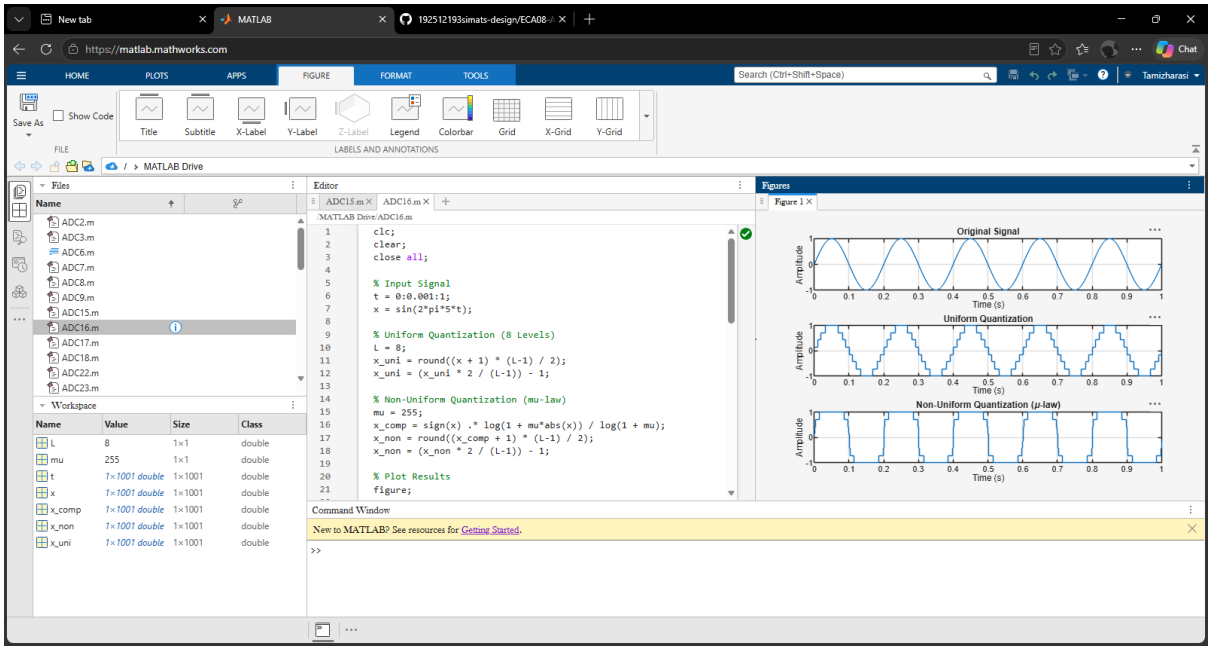
EXP9


clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;
% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);
% AM Modulation
am = (1 + m) .* cos(2*pi*fc*t);
% Add AWGN manually
SNR_dB = 10;
signalPower = mean(am.^2);
noisePower = signalPower / 10^(SNR_dB/10);
Message Signal
Carrier Signal
AM Signal
AM Signal with AWGN Noise
Recovered Signal
Amplitude
Time (s)

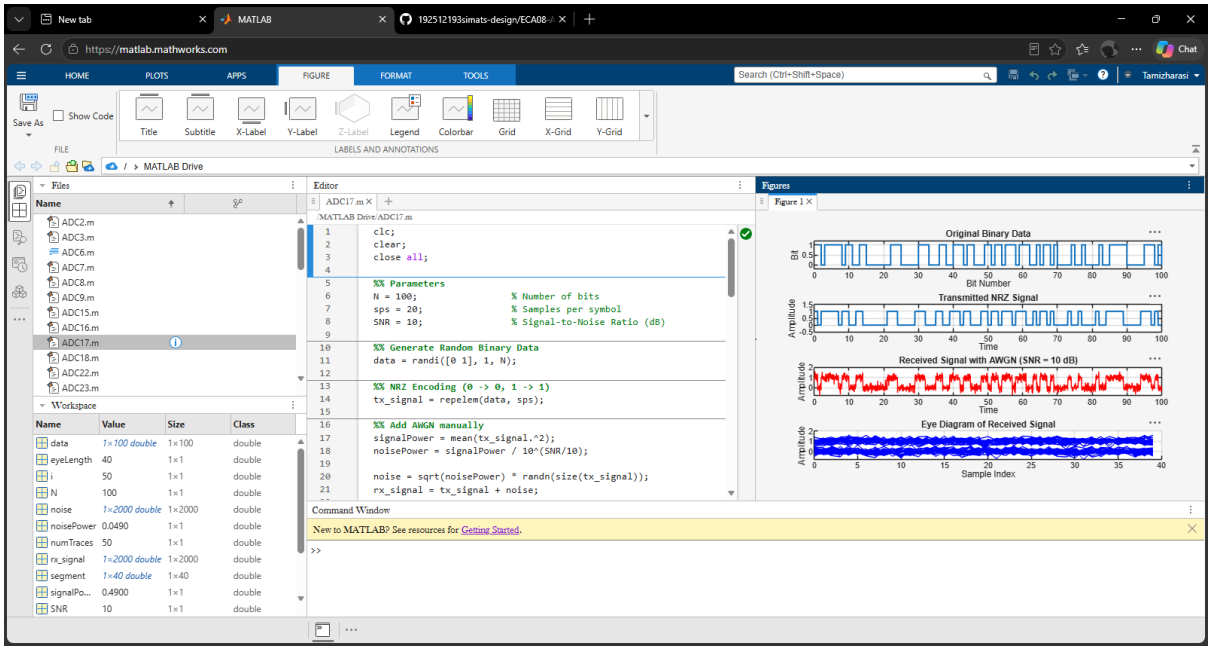
EXP 15


clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500; % Message frequency
fc = 5000; % Carrier frequency
kf = 100; % Frequency sensitivity
% Message and Carrier Signals
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);
% FM Modulation
fm_sig = cos(2*pi*fc*t + 2*pi*kf*cumsum(m)/fs);
% Add Gaussian White Noise manually
SNR_dB = 10;
signalPower = mean(fm_sig.^2);
Message Signal
Carrier Signal
FM Signal
FM Signal with Gaussian White Noise
Amplitude
Time (s)

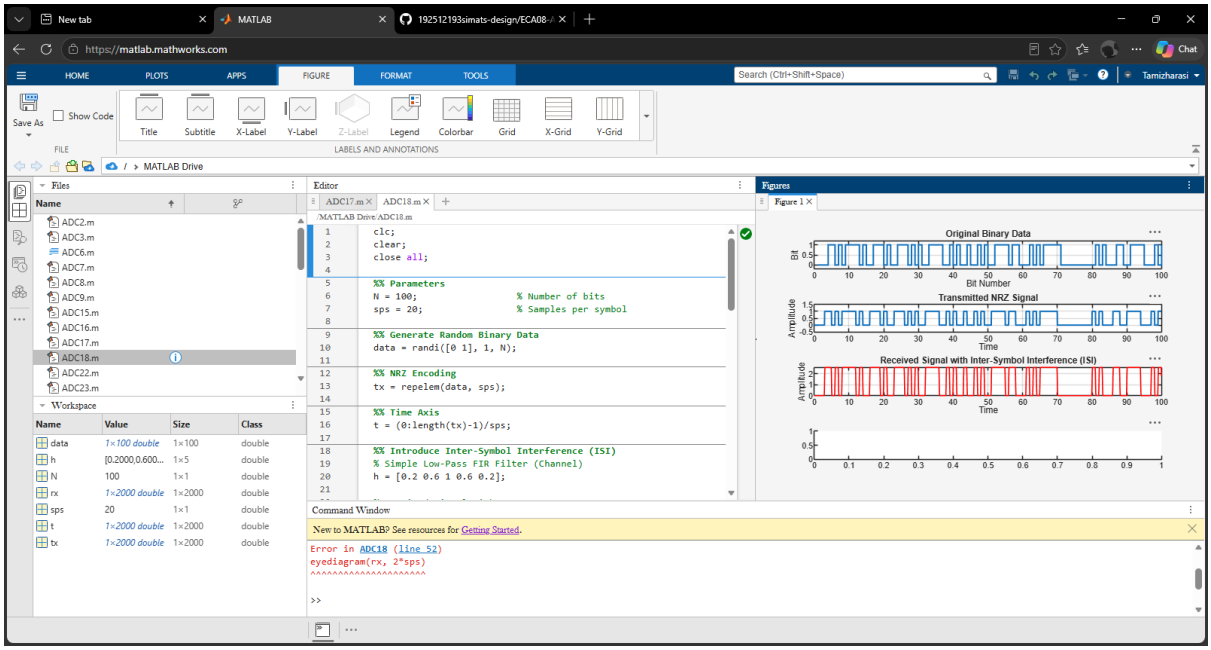
EXP16


clc;
clear;
close all;
% Input Signal
t = 0:0.001:1;
x = sin(2*pi*5*t);
% Uniform Quantization (8 Levels)
L = 8;
x_uni = round((x + 1) * (L-1) / 2);
x_uni = (x_uni * 2 / (L-1)) - 1;
% Non-Uniform Quantization (mu-law)
mu = 255;
x_comp = sign(x) .* log(1 + mu*abs(x)) / log(1 + mu);
x_non = round((x_comp + 1) * (L-1) / 2);
x_non = (x_non * 2 / (L-1)) - 1;
% Plot Results
figure;
Original Signal
Uniform Quantization
Non-Uniform Quantization (μ-law)
Amplitude
Time (s)

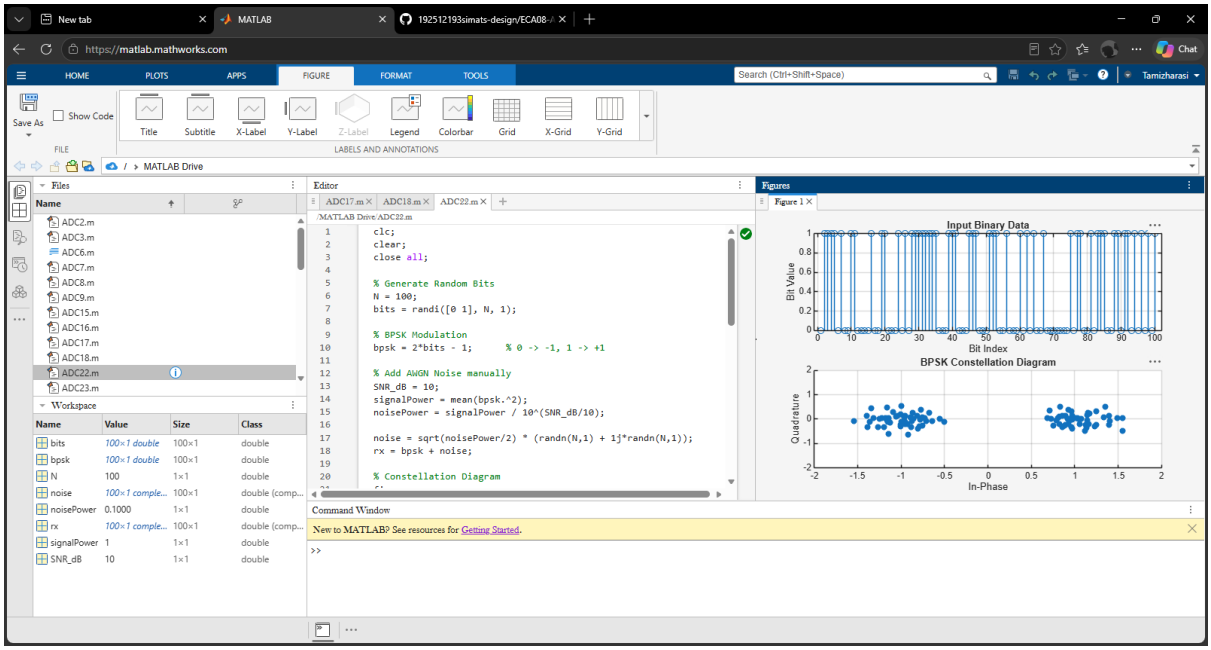
EXP17


clc;
clear;
close all;
%% Parameters
N = 100; % Number of bits
sps = 20; % Samples per symbol
SNR = 10; % Signal-to-Noise Ratio (dB)
%% Generate Random Binary Data
data = randi([0 1], 1, N);
%% NRZ Encoding (0 -> 0, 1 -> 1)
tx_signal = repelem(data, sps);
%% Add AWGN manually
signalPower = mean(tx_signal.^2);
noisePower = signalPower / 10^(SNR/10);
noise = sqrt(noisePower) * randn(size(tx_signal));
rx_signal = tx_signal + noise;
Original Binary Data
Transmitted NRZ Signal
Received Signal with AWGN (SNR = 10 dB)
Eye Diagram of Received Signal
Bit Number
Time
Sample Index

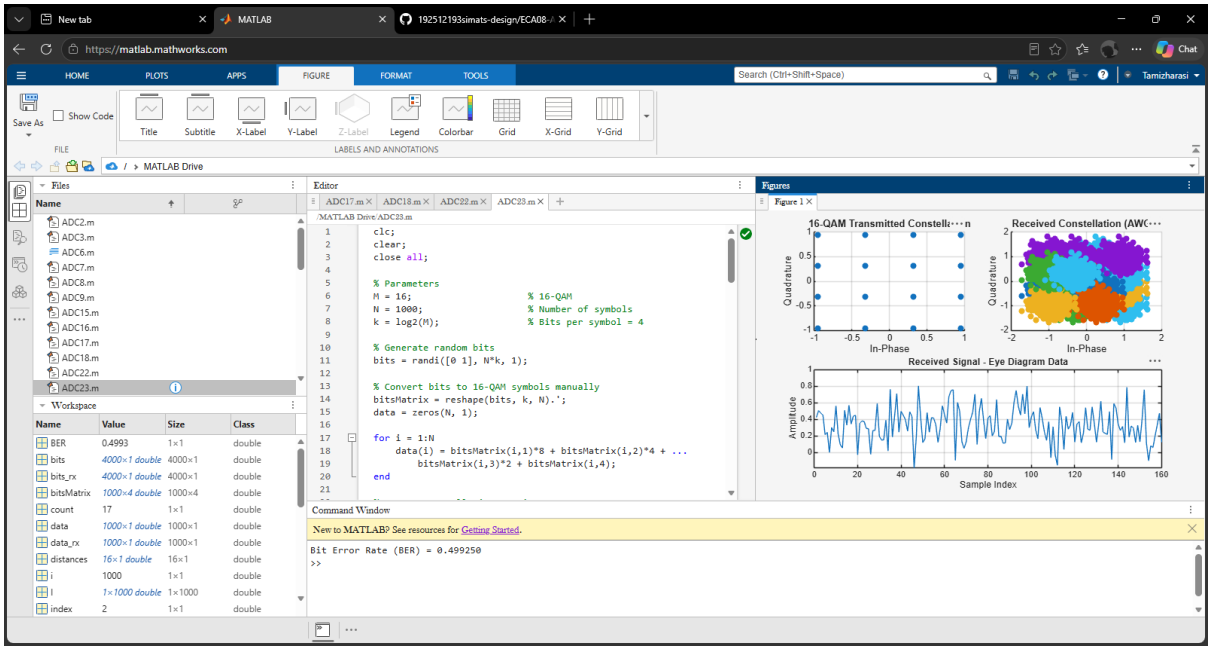
EXP18


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ADC9.m
ADC15.m
ADC16.m
ADC17.m
ADC18.m
ADC22.m
ADC23.m
Workspace
Name
Value
Size
Class
data
1×100 double
1×100
double
h
[0.2000,0.600...
1×5
double
N
100
1×1
double
rx
1×2000 double
1×2000
double
sps
20
1×1
double
t
1×2000 double
1×2000
double
tx
1×2000 double
1×2000
double
Editor
ADC17.m
ADC18.m
/MATLAB Drive/ADC18.m
clc;
clear;
close all;
%% Parameters
N = 100;
% Number of bits
sps = 20;
% Samples per symbol
%% Generate Random Binary Data
data = randi([0 1], 1, N);
%% NRZ Encoding
tx = repelem(data, sps);
%% Time Axis
t = (0:length(tx)-1)/sps;
%% Introduce Inter-Symbol Interference (ISI)
% Simple Low-Pass FIR Filter (Channel)
h = [0.2 0.6 1 0.6 0.2];
Figures
Figure 1
Original Binary Data
Bit
Bit Number
Transmitted NRZ Signal
Amplitude
Time
Received Signal with Inter-Symbol Interference (ISI)
Amplitude
Time
Command Window
New to MATLAB? See resources for Getting Started.
Error in ADC18 (line 52)
eyediagram(rx, 2*sps)
^^^^^^^^^^^^^^^^^^^^
>>

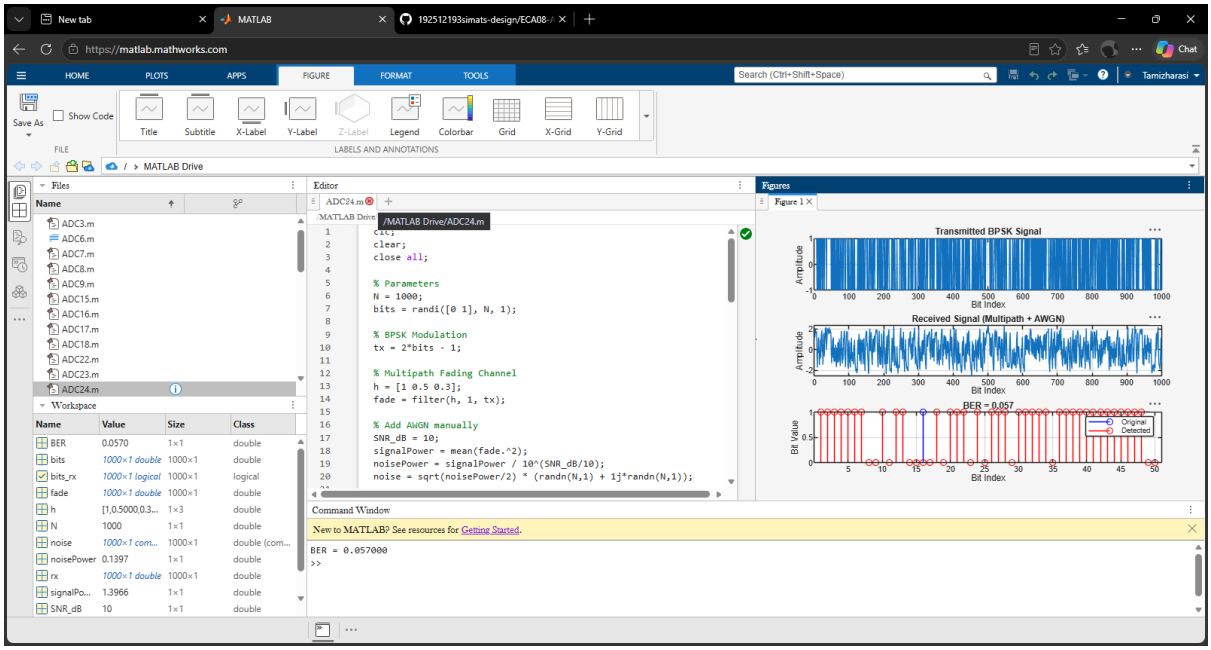
EXP22


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ADC16.m
ADC17.m
ADC18.m
ADC22.m
ADC23.m
Workspace
Name
Value
Size
Class
bits
100×1 double
100×1
double
bpsk
100×1 double
100×1
double
N
100
1×1
double
noise
100×1 comple...
100×1
double (comp...
noisePower
0.1000
1×1
double
rx
100×1 comple...
100×1
double (comp...
signalPower
1
1×1
double
SNR_dB
10
1×1
double
Editor
ADC17.m
ADC18.m
ADC22.m
/MATLAB Drive/ADC22.m
clc;
clear;
close all;
% Generate Random Bits
N = 100;
bits = randi([0 1], N, 1);
% BPSK Modulation
bpsk = 2*bits - 1;
% 0 -> -1, 1 -> +1
% Add AWGN Noise manually
SNR_dB = 10;
signalPower = mean(bpsk.^2);
noisePower = signalPower / 10^(SNR_dB/10);
noise = sqrt(noisePower/2) * (randn(N,1) + 1j*randn(N,1));
rx = bpsk + noise;
% Constellation Diagram
Figures
Figure 1
Input Binary Data
Bit Value
Bit Index
BPSK Constellation Diagram
Quadrature
In-Phase
Command Window
New to MATLAB? See resources for Getting Started.
>>

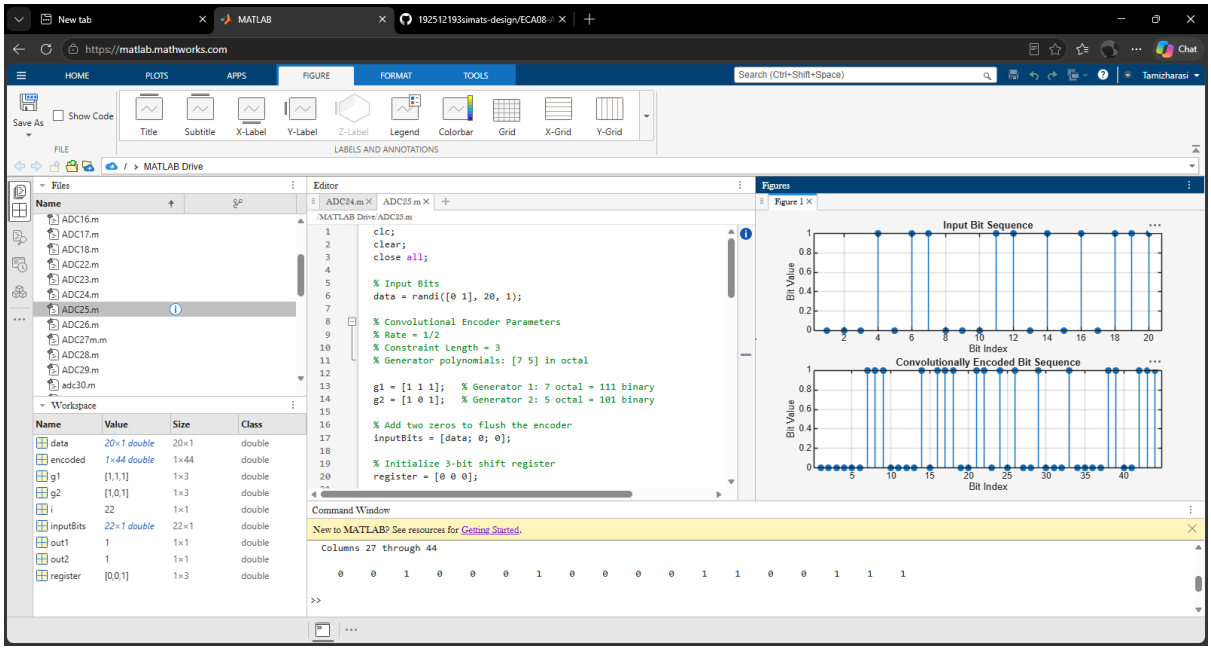
EXP23


16-QAM Transmitted Constellation
Received Constellation (AWGN)
In-Phase
Quadrature
Received Signal - Eye Diagram Data
Amplitude
Sample Index
Command Window
Bit Error Rate (BER) = 0.499250

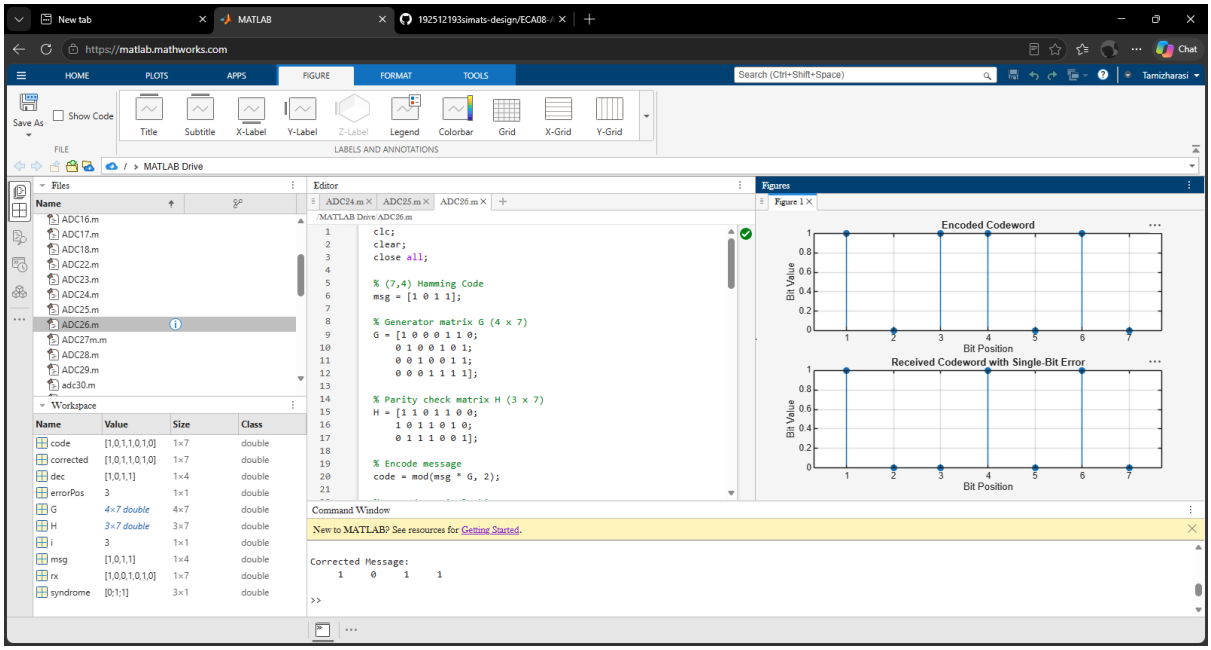
EXP24


Transmitted BPSK Signal
Received Signal (Multipath + AWGN)
BER = 0.057
Original
Detected
Bit Index
Amplitude
Bit Value
Command Window
BER = 0.057000

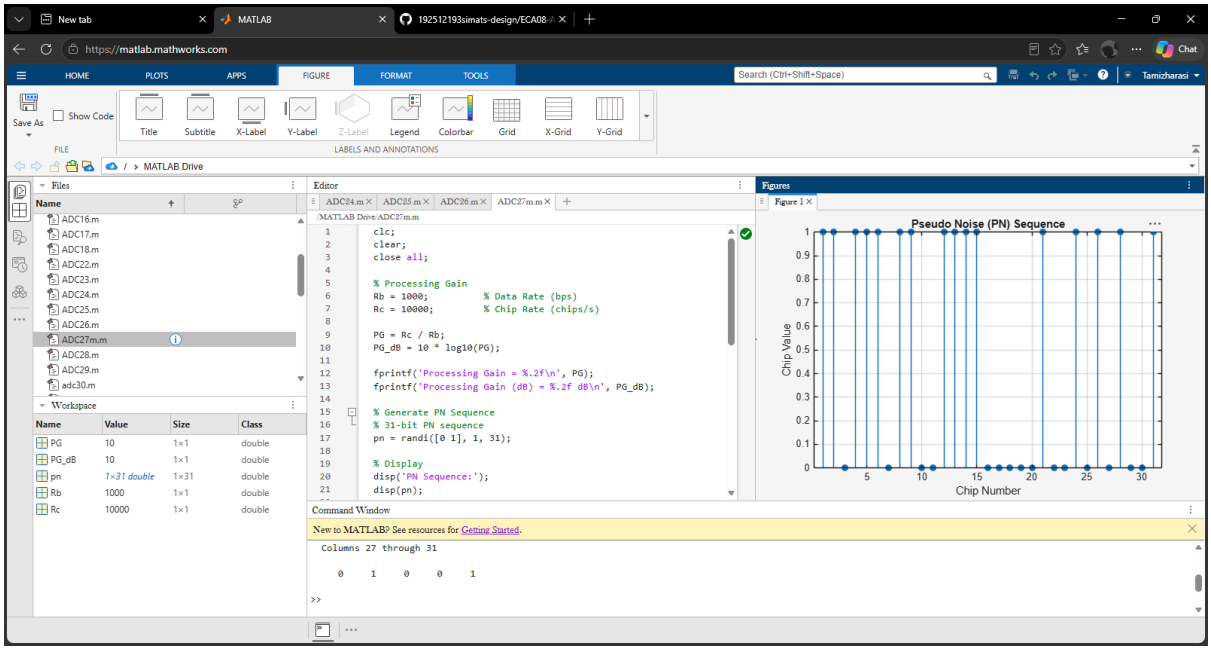
EXP25


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ADC18.m
ADC22.m
ADC23.m
ADC24.m
ADC25.m
ADC26.m
ADC27m.m
ADC28.m
ADC29.m
adc30.m
Workspace
Name
Value
Size
Class
data
20×1 double
20×1
double
encoded
1×44 double
1×44
double
g1
[1,1,1]
1×3
double
g2
[1,0,1]
1×3
double
i
22
1×1
double
inputBits
22×1 double
22×1
double
out1
1
1×1
double
out2
1
1×1
double
register
[0,0,1]
1×3
double
Editor
ADC24.m
ADC25.m
MATLAB Drive/ADC25.m
clc;
clear;
close all;
% Input Bits
data = randi([0 1], 20, 1);
% Convolutional Encoder Parameters
% Rate = 1/2
% Constraint Length = 3
% Generator polynomials: [7 5] in octal
g1 = [1 1 1]; % Generator 1: 7 octal = 111 binary
g2 = [1 0 1]; % Generator 2: 5 octal = 101 binary
% Add two zeros to flush the encoder
inputBits = [data; 0; 0];
% Initialize 3-bit shift register
register = [0 0 0];
Figures
Figure 1
Input Bit Sequence
Bit Value
Bit Index
Convolutionally Encoded Bit Sequence
Bit Value
Bit Index
Command Window
New to MATLAB? See resources for Getting Started.
Columns 27 through 44
0 0 1 0 0 0 1 0 0 0 0 1 1 0 0 1 1 1
>>

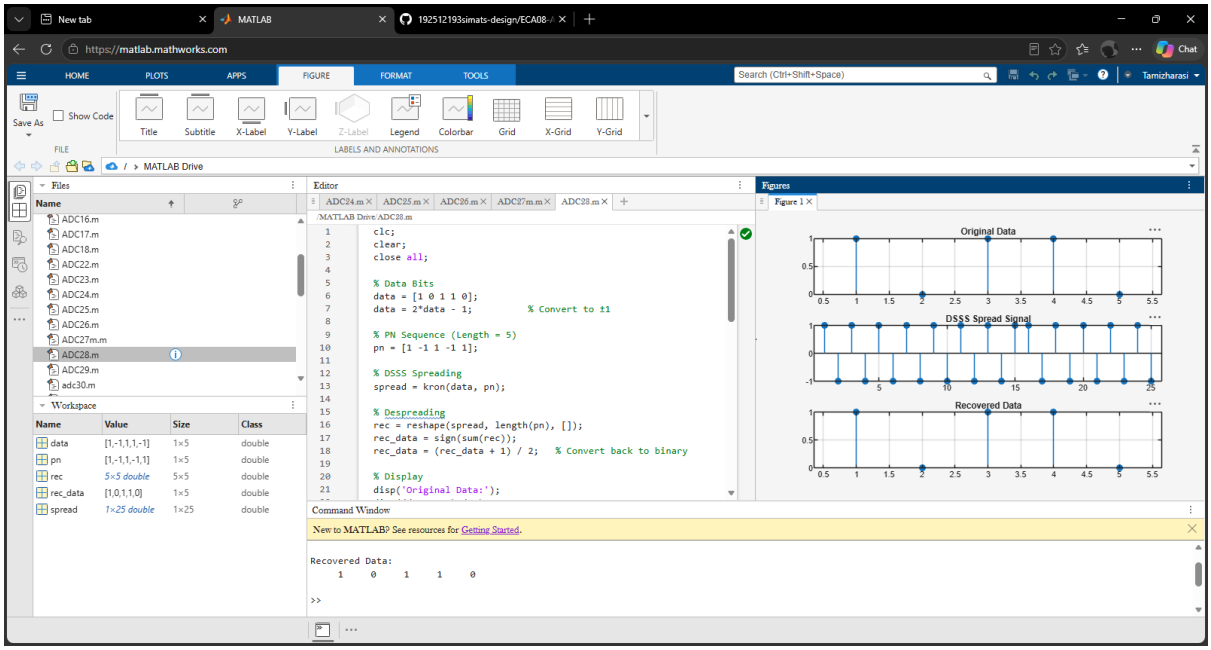
EXP26


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ADC17.m
ADC18.m
ADC22.m
ADC23.m
ADC24.m
ADC25.m
ADC26.m
ADC27m.m
ADC28.m
ADC29.m
adc30.m
Workspace
Name
Value
Size
Class
code
[1,0,1,1,0,1,0]
1×7
double
corrected
[1,0,1,1,0,1,0]
1×7
double
dec
[1,0,1,1]
1×4
double
errorPos
3
1×1
double
G
4×7 double
4×7
double
H
3×7 double
3×7
double
i
3
1×1
double
msg
[1,0,1,1]
1×4
double
rx
[1,0,0,1,0,1,0]
1×7
double
syndrome
[0;1;1]
3×1
double
Editor
ADC24.m
ADC25.m
ADC26.m
MATLAB Drive/ADC26.m
clc;
clear;
close all;
% (7,4) Hamming Code
msg = [1 0 1 1];
% Generator matrix G (4 x 7)
G = [1 0 0 0 1 1 0;
0 1 0 0 1 0 1;
0 0 1 0 0 1 1;
0 0 0 1 1 1 1];
% Parity check matrix H (3 x 7)
H = [1 1 0 1 1 0 0;
1 0 1 1 0 1 0;
0 1 1 1 0 0 1];
% Encode message
code = mod(msg * G, 2);
Figures
Figure 1
Encoded Codeword
Bit Value
Bit Position
Received Codeword with Single-Bit Error
Bit Value
Bit Position
Command Window
New to MATLAB? See resources for Getting Started.
Corrected Message:
1 0 1 1
>>

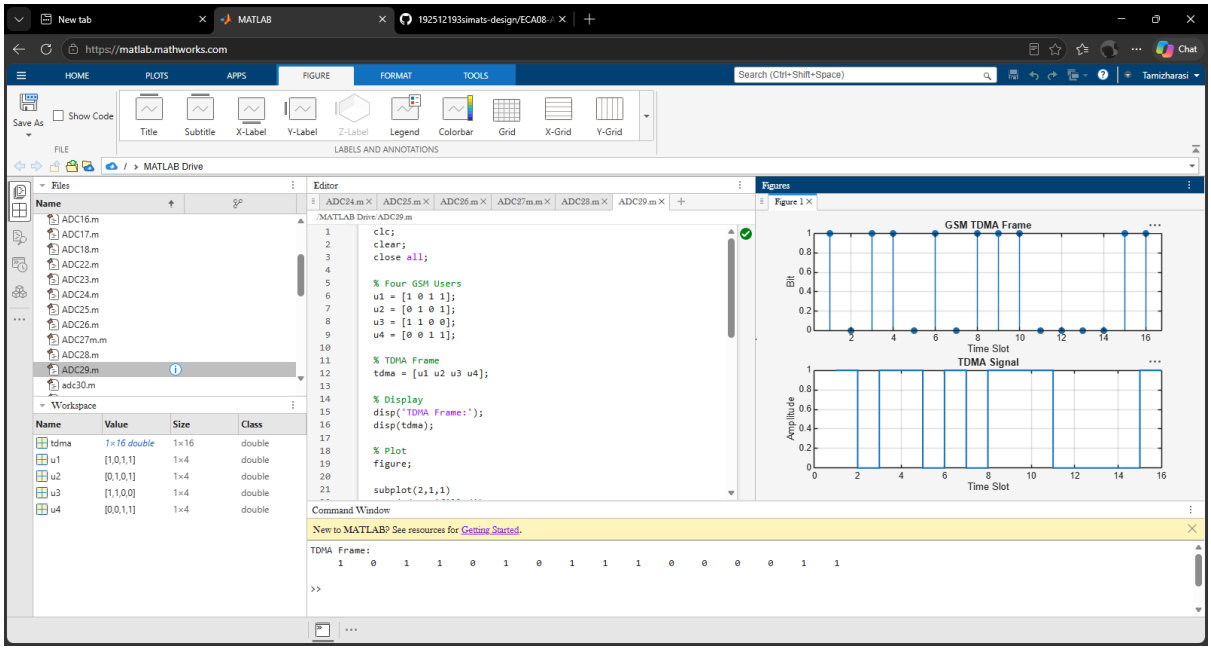
EXP27


Pseudo Noise (PN) Sequence
Chip Value
Chip Number
clc;
clear;
close all;
% Processing Gain
Rb = 1000; % Data Rate (bps)
Rc = 10000; % Chip Rate (chips/s)
PG = Rc / Rb;
PG_dB = 10 * log10(PG);
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);
% Generate PN Sequence
% 31-bit PN sequence
pn = randi([0 1], 1, 31);
% Display
disp('PN Sequence:');
disp(pn);
Columns 27 through 31
0 1 0 0 1

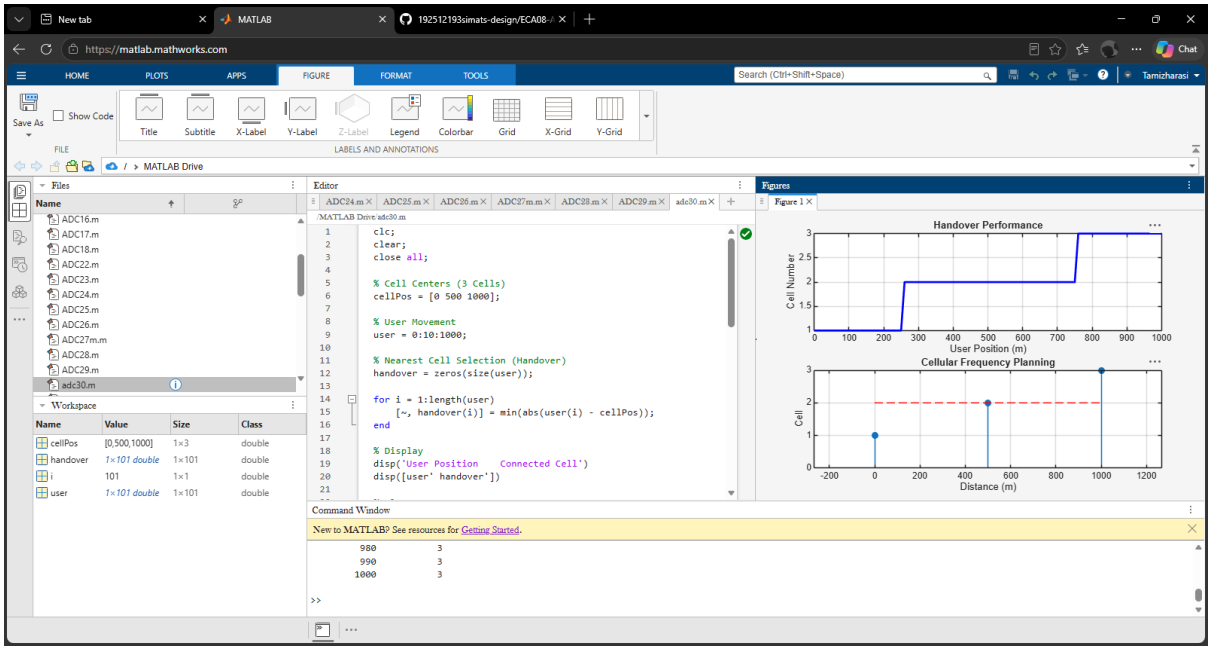
EXP28


Original Data
DSSS Spread Signal
Recovered Data
clc;
clear;
close all;
% Data Bits
data = [1 0 1 1 0];
data = 2*data - 1; % Convert to ±1
% PN Sequence (Length = 5)
pn = [1 -1 1 -1 1];
% DSSS Spreading
spread = kron(data, pn);
% Despreading
rec = reshape(spread, length(pn), []);
rec_data = sign(sum(rec));
rec_data = (rec_data + 1) / 2; % Convert back to binary
% Display
disp('Original Data:');
Recovered Data:
1 0 1 1 0

EXP29


GSM TDMA Frame
Bit
Time Slot
TDMA Signal
Amplitude
Time Slot
TDMA Frame:
1 0 1 1 0 1 0 1 1 1 0 0 0 0 1 1

EXP30


Handover Performance
Cell Number
User Position (m)
Cellular Frequency Planning
Cell
Distance (m)
980 3
990 3
1000 3